

► BIOCHEMISTRY—METABOLISM

Enzyme terminology	An enzyme's name often describes its function. For example, glucokinase is an enzyme that catalyzes the phosphorylation of glucose using a molecule of ATP. The following are commonly used enzyme descriptors.
Kinase	Catalyzes transfer of a phosphate group from a high-energy molecule (usually ATP) to a substrate (eg, phosphofructokinase).
Phosphorylase	Adds inorganic phosphate onto substrate without using ATP (eg, glycogen phosphorylase).
Phosphatase	Removes phosphate group from substrate (eg, fructose-1,6-bisphosphatase 1).
Dehydrogenase	Catalyzes oxidation-reduction reactions (eg, pyruvate dehydrogenase).
Hydroxylase	Adds hydroxyl group (–OH) onto substrate (eg, tyrosine hydroxylase).
Carboxylase	Transfers carboxyl groups (–COOH) with the help of biotin (eg, pyruvate carboxylase).
Mutase	Relocates a functional group within a molecule (eg, vitamin B ₁₂ –dependent methylmalonyl-CoA mutase).
Synthase	Catalyzes synthesis reactions without using ATP as a source of energy.

Rate-determining enzymes of metabolic processes

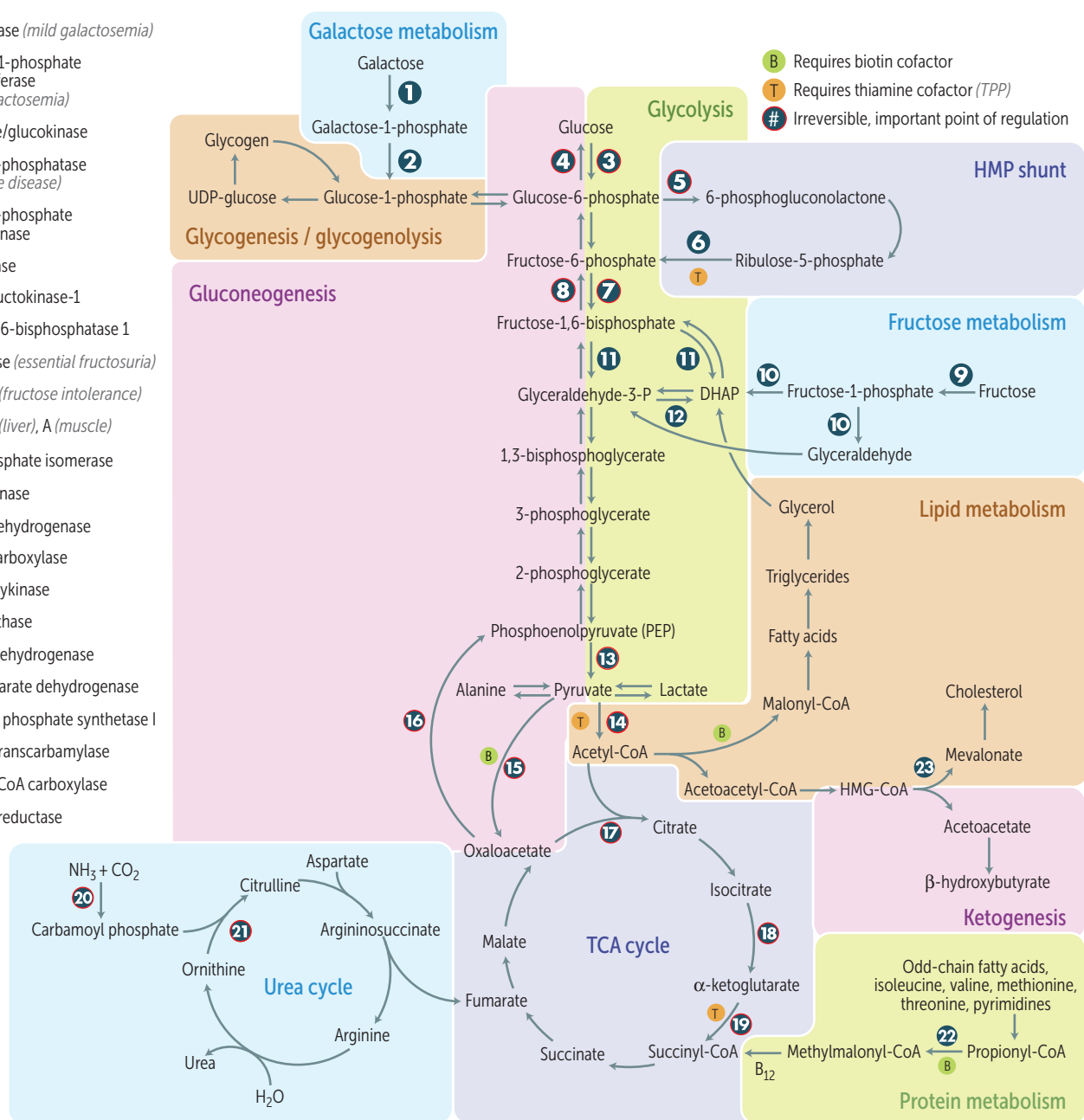
PROCESS	ENZYME	REGULATORS
Glycolysis	Phosphofructokinase-1 (PFK-1)	AMP ⊕, fructose-2,6-bisphosphate ⊕ ATP ⊖, citrate ⊖
Gluconeogenesis	Fructose-1,6-bisphosphatase 1	AMP ⊖, fructose-2,6-bisphosphate ⊖
TCA cycle	Isocitrate dehydrogenase	ADP ⊕ ATP ⊖, NADH ⊖
Glycogenesis	Glycogen synthase	Glucose-6-phosphate ⊕, insulin ⊕, cortisol ⊕ Epinephrine ⊖, glucagon ⊖
Glycogenolysis	Glycogen phosphorylase	Epinephrine ⊕, glucagon ⊕, AMP ⊕ Glucose-6-phosphate ⊖, insulin ⊖, ATP ⊖
HMP shunt	Glucose-6-phosphate dehydrogenase (G6PD)	NADP ⁺ ⊕ NADPH ⊖
De novo pyrimidine synthesis	Carbamoyl phosphate synthetase II	ATP ⊕, PRPP ⊕ UTP ⊖
De novo purine synthesis	Glutamine-phosphoribosylpyrophosphate (PRPP) amidotransferase	PRPP ⊕, AMP ⊖, inosine monophosphate (IMP) ⊖, GMP ⊖
Urea cycle	Carbamoyl phosphate synthetase I	N-acetylglutamate ⊕
Fatty acid synthesis	Acetyl-CoA carboxylase (ACC)	Insulin ⊕, citrate ⊕ Glucagon ⊖, palmitoyl-CoA ⊖
Fatty acid oxidation	Carnitine acyltransferase I	Malonyl-CoA ⊖
Ketogenesis	HMG -CoA synthase (HOMG ! I'm starving!)	
Cholesterol synthesis	HMG-CoA reductase	Insulin ⊕, thyroxine ⊕, estrogen ⊕ Glucagon ⊖, cholesterol ⊖

Metabolic compartmentation

Mitochondria	Fatty acid oxidation (β -oxidation), acetyl-CoA production, TCA cycle, oxidative phosphorylation, ketogenesis.
Cytoplasm	Glycolysis, HMP shunt, and synthesis of cholesterol (SER), proteins (ribosomes, RER), fatty acids, and nucleotides.
Both	Heme synthesis, urea cycle, gluconeogenesis. Hugs take two (both).

Summary of pathways

- 1 Galactokinase (*mild galactosemia*)
- 2 Galactose-1-phosphate uridylyltransferase (*severe galactosemia*)
- 3 Hexokinase/glucokinase
- 4 Glucose-6-phosphatase (*von Gierke disease*)
- 5 Glucose-6-phosphate dehydrogenase
- 6 Transketolase
- 7 Phosphofructokinase-1
- 8 Fructose-1,6-bisphosphatase 1
- 9 Fructokinase (*essential fructosuria*)
- 10 Aldolase B (*fructose intolerance*)
- 11 Aldolase B (*liver*), A (*muscle*)
- 12 Triose phosphate isomerase
- 13 Pyruvate kinase
- 14 Pyruvate dehydrogenase
- 15 Pyruvate carboxylase
- 16 PEP carboxykinase
- 17 Citrate synthase
- 18 Isocitrate dehydrogenase
- 19 α -ketoglutarate dehydrogenase
- 20 Carbamoyl phosphate synthetase I
- 21 Ornithine transcarbamylase
- 22 Propionyl-CoA carboxylase
- 23 HMG-CoA reductase



Activated carriers

CARRIER MOLECULE	CARRIED IN ACTIVATED FORM
ATP	Phosphoryl groups
NADH, NADPH, FADH ₂	Electrons
CoA, lipoamide	Acyl groups
Biotin	CO ₂
Tetrahydrofolates	1-carbon units
S-adenosylmethionine (SAM)	CH ₃ groups
TPP	Aldehydes

Universal electron acceptors

Nicotinamides (NAD ⁺ , NADP ⁺ from vitamin B ₃) and flavin nucleotides (FAD from vitamin B ₂). NAD ⁺ is generally used in catabolic processes to carry reducing equivalents away as NADH. NADPH is used in anabolic processes (eg, steroid and fatty acid synthesis) as a supply of reducing equivalents.	NADPH is a product of the HMP shunt. NADPH is used in: ▪ Anabolic processes ▪ Respiratory burst ▪ Cytochrome P-450 system ▪ Glutathione reductase
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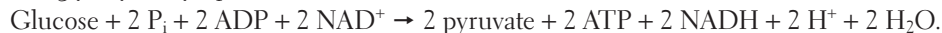
Hexokinase vs glucokinase

Phosphorylation of glucose to yield glucose-6-phosphate is catalyzed by glucokinase in the liver and hexokinase in other tissues. Hexokinase sequesters glucose in tissues, where it is used even when glucose concentrations are low. At high glucose concentrations, glucokinase helps to store glucose in liver. Glucokinase deficiency (→ ↑↑ glucose needed for activation → impaired insulin release [vs. diabetes mellitus]) is a cause of maturity onset diabetes of the young (MODY) and gestational diabetes.

	Hexokinase	Glucokinase
Location	Most tissues, except liver and pancreatic β cells	Liver, β cells of pancreas
K _m	Lower (↑ affinity)	Higher (↓ affinity)
V _{max}	Lower (↓ capacity)	Higher (↑ capacity)
Induced by insulin	No	Yes
Feedback inhibition by	Glucose-6-phosphate	Fructose-6-phosphate

Glycolysis regulation, key enzymes

Net glycolysis (cytoplasm):

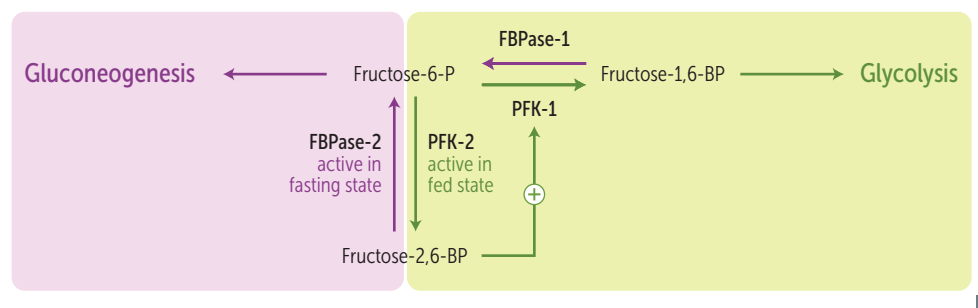


Equation not balanced chemically, and exact balanced equation depends on ionization state of reactants and products.

REQUIRE ATP	Glucose $\xrightarrow{\text{Hexokinase/glucokinase}}$ Glucose-6-P	Glucose-6-P $\ominus$ hexokinase. Fructose-6-P $\ominus$ glucokinase.
	Fructose-6-P $\xrightarrow[\text{(rate-limiting step)}]{\text{Phosphofructokinase-1}}$ Fructose-1,6-BP	AMP $\oplus$, fructose-2,6-bisphosphate $\oplus$. ATP $\ominus$, citrate $\ominus$.
PRODUCE ATP	1,3-BPG $\xleftarrow{\text{Phosphoglycerate kinase}}$ 3-PG	
	Phosphoenolpyruvate $\xrightarrow{\text{Pyruvate kinase}}$ Pyruvate	Fructose-1,6-bisphosphate $\oplus$. ATP $\ominus$, alanine $\ominus$, glucagon $\ominus$, epinephrine $\ominus$.

Regulation by fructose-2,6-bisphosphate

Fructose biphosphatase-2 (FBPase-2) and phosphofructokinase-2 (PFK-2) are the same bifunctional enzyme whose function is reversed by phosphorylation by protein kinase A.



Fasting state: $\uparrow$ glucagon $\rightarrow$ $\uparrow$ cAMP $\rightarrow$ $\uparrow$ protein kinase A $\rightarrow$ $\uparrow$ FBPase-2, $\downarrow$ PFK-2, less glycolysis, more gluconeogenesis.

FaBian the **P**easant (**FBP**) has to work hard when starving.

Fed state: $\uparrow$ insulin $\rightarrow$ $\uparrow$ PFK-2 $\rightarrow$ more glycolysis, less gluconeogenesis.

Prince Frederic**K** (**PFK**) works only when fed.

Pyruvate dehydrogenase complex

Mitochondrial enzyme complex linking glycolysis and TCA cycle. Differentially regulated in fed (active)/fasting (inactive) states.
Reaction: pyruvate + NAD^+ + CoA $\rightarrow$ acetyl-CoA + CO_2 + NADH.

Contains 3 enzymes requiring 5 cofactors:

1. **T**hiamine pyrophosphate (B_1)
2. **L**ipoic acid
3. **C**oA (B_5 , pantothenic acid)
4. **FAD** (B_2 , riboflavin)
5. **NAD** $^+$ (B_3 , niacin)

Activated by: $\uparrow$ NAD^+/NADH ratio, $\uparrow$ ADP
 $\uparrow$ Ca^{2+} .

The complex is similar to the α -ketoglutarate dehydrogenase complex (same cofactors, similar substrate and action), which converts α -ketoglutarate $\rightarrow$ succinyl-CoA (TCA cycle).

The **l**ovely **c**oenzymes **f**or **n**erds.

Arsenic inhibits lipoic acid. Arsenic poisoning clinical findings: imagine a vampire (pigmentary skin changes, skin cancer), vomiting and having diarrhea, running away from a cutie (**Q**T prolongation) with garlic breath.

Pyruvate dehydrogenase complex deficiency

Causes a buildup of pyruvate that gets shunted to lactate (via LDH) and alanine (via ALT).
X-linked.

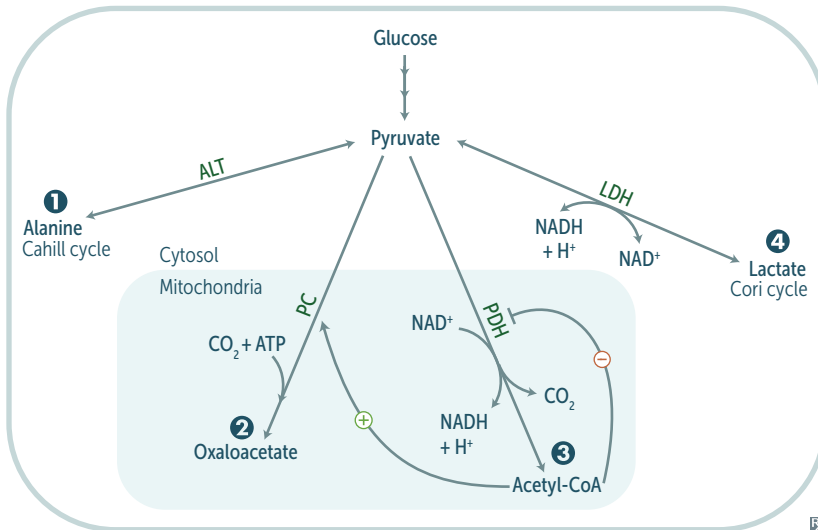
FINDINGS

Neurologic defects, lactic acidosis, ↑ serum alanine starting in infancy.

TREATMENT

↑ intake of ketogenic nutrients (eg, high fat content or ↑ lysine and leucine), B₁ and lipoic acid.

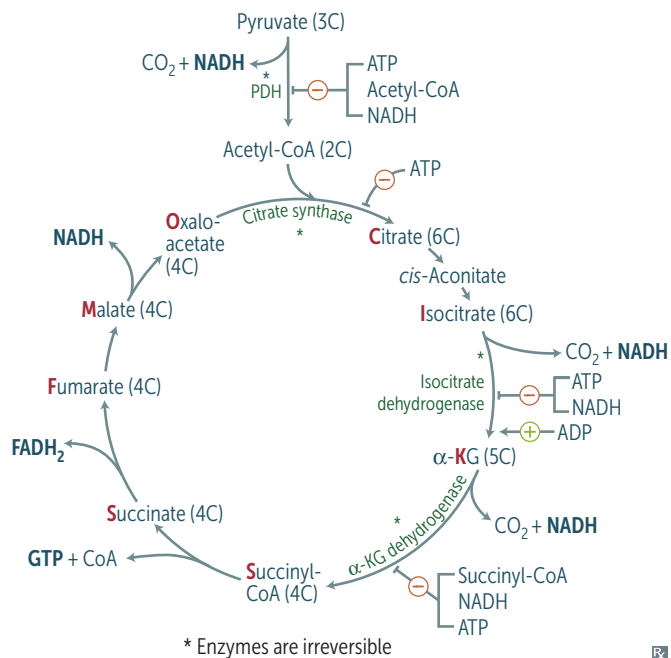
Pyruvate metabolism



Functions of different pyruvate metabolic pathways (and their associated cofactors):

- 1 Alanine aminotransferase (B₆): alanine carries amino groups to the liver from muscle
- 2 Pyruvate carboxylase (B₇): oxaloacetate can replenish TCA cycle or be used in gluconeogenesis
- 3 Pyruvate dehydrogenase (B₁, B₂, B₃, B₅, lipoic acid): transition from glycolysis to the TCA cycle
- 4 Lactic acid dehydrogenase (B₃): end of anaerobic glycolysis (major pathway in RBCs, WBCs, kidney medulla, lens, cornea, and Sertoli cells in testes)

TCA cycle



Also called Krebs cycle. Pyruvate → acetyl-CoA produces 1 NADH, 1 CO₂.

The TCA cycle produces 3 NADH, 1 FADH₂, 2 CO₂, 1 GTP per acetyl-CoA = 10 ATP/ acetyl-CoA (2× everything per glucose). TCA cycle reactions occur in the mitochondria.

α-ketoglutarate dehydrogenase complex requires the same cofactors as the pyruvate dehydrogenase complex (vitamins B₁, B₂, B₃, B₅, lipoic acid).

Citrate is Krebs' starting substrate for making oxaloacetate.

Electron transport chain and oxidative phosphorylation

NADH electrons are transferred to complex I.
FADH₂ electrons are transferred to complex II (at a lower energy level than NADH).

Oxygen acts as an electron acceptor to provide energy.

The passage of electrons results in the formation of a proton gradient that, coupled to oxidative phosphorylation, drives ATP production. ATP hydrolysis can be coupled to energetically unfavorable reactions.

Uncoupling proteins (found in brown fat, which has more mitochondria than white fat) produce heat by ↑ inner mitochondrial membrane permeability → ↓ proton gradient. ATP synthesis stops, but electron transport continues.

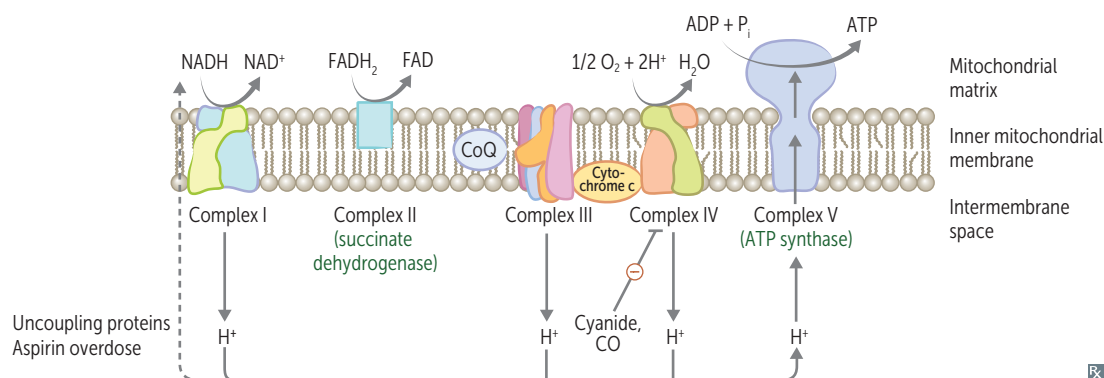
1 NADH → 2.5 ATP; 1 FADH₂ → 1.5 ATP

NADH electrons from glycolysis enter mitochondria via the malate-aspartate or glycerol-3-phosphate shuttle.

Aerobic metabolism of one glucose molecule produces 32 net ATP via malate-aspartate shuttle (heart and liver), 30 net ATP via glycerol-3-phosphate shuttle (muscle).

Anaerobic glycolysis produces only 2 net ATP per glucose molecule.

Aspirin overdose can also cause uncoupling of oxidative phosphorylation resulting in hyperthermia.



Gluconeogenesis, irreversible enzymes

All enzymes may be subject to activation by glucagon in fasting state.

Pathway produces fresh glucose.

Pyruvate carboxylase

In mitochondria. Pyruvate → oxaloacetate.

Requires biotin, ATP. Activated by acetyl-CoA.

Phosphoenolpyruvate carboxykinase

In cytosol. Oxaloacetate → phosphoenolpyruvate (PEP).

Requires GTP.

Fructose-1,6-bisphosphatase 1

In cytosol. Fructose-1,6-bisphosphate → fructose-6-phosphate.

Citrate ⊕, AMP ⊖, fructose 2,6-bisphosphate ⊖.

Glucose-6-phosphatase

In ER. Glucose-6-phosphate → glucose.

Occurs primarily in liver; serves to maintain euglycemia during fasting. Enzymes also found in kidney, intestinal epithelium. Deficiency of the key gluconeogenic enzymes causes hypoglycemia. (Muscle cannot participate in gluconeogenesis because it lacks glucose-6-phosphatase).

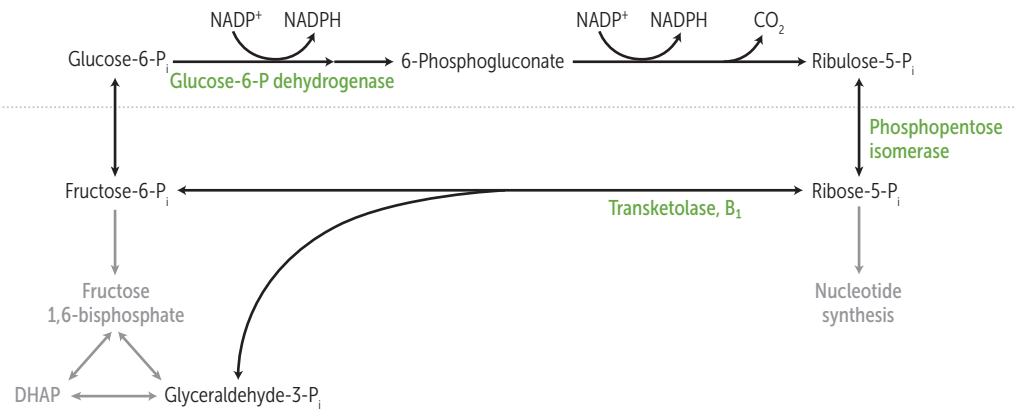
Odd-chain fatty acids yield 1 propionyl-CoA during metabolism, which can enter the TCA cycle (as succinyl-CoA), undergo gluconeogenesis, and serve as a glucose source (It's odd for fatty acids to make glucose). Even-chain fatty acids cannot produce new glucose, since they yield only acetyl-CoA equivalents.

Pentose phosphate pathway

Also called HMP shunt. Provides a source of NADPH from abundantly available glucose-6-P (NADPH is required for reductive reactions, eg, glutathione reduction inside RBCs, fatty acid and cholesterol biosynthesis). Additionally, this pathway yields ribose for nucleotide synthesis. Two distinct phases (oxidative and nonoxidative), both of which occur in the cytoplasm. No ATP is used or produced.

Sites: lactating mammary glands, liver, adrenal cortex (sites of fatty acid or steroid synthesis), RBCs.

REACTIONS

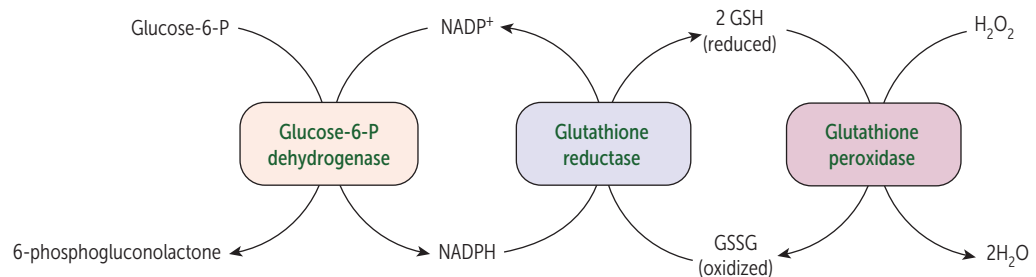
**Oxidative
(irreversible)****Nonoxidative
(reversible)****Glucose-6-phosphate dehydrogenase deficiency**

NADPH is necessary to keep glutathione reduced, which in turn detoxifies free radicals and peroxides. ↓ NADPH in RBCs leads to hemolytic anemia due to poor RBC defense against oxidizing agents (eg, fava beans, sulfonamides, nitrofurantoin, primaquine). Infection (most common cause) can also precipitate hemolysis; inflammatory response produces free radicals that diffuse into RBCs, causing oxidative damage.

X-linked recessive disorder; most common human enzyme deficiency; more prevalent among descendants of populations in malaria-endemic regions (eg, sub-Saharan Africa, Southeast Asia).

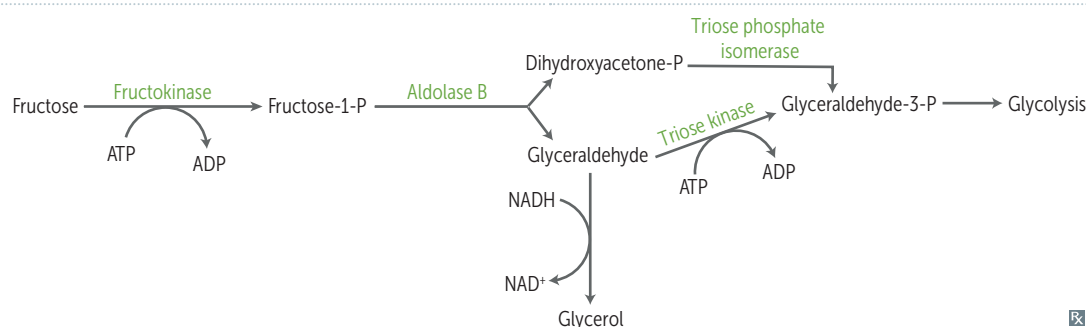
Heinz bodies—denatured globin chains precipitate within RBCs due to oxidative stress.

Bite cells—result from the phagocytic removal of **Heinz** bodies by splenic macrophages. Think, “**Bite** into some **Heinz** ketchup.”



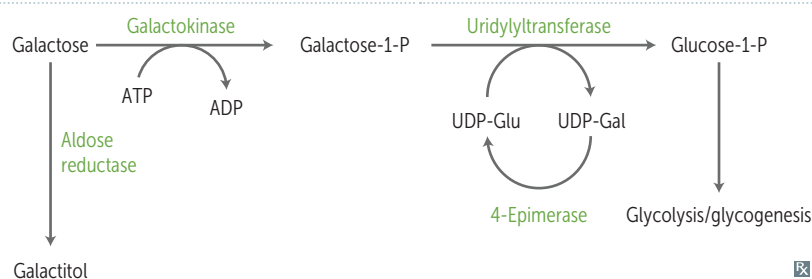
Disorders of fructose metabolism

	Essential fructosuria	Hereditary fructose intolerance
ENZYME DEFICIENCY	Fructokinase (autosomal recessive)	Aldolase B (autosomal recessive)
PATHOPHYSIOLOGY	Fructose is not trapped into cells. Hexokinase becomes 1° pathway for converting fructose to fructose-1-phosphate.	Fructose-1-phosphate accumulates → ↓ available phosphate → inhibition of glycogenolysis and gluconeogenesis.
PRESENTATION (SIGNS/SYMPTOMS)	Asymptomatic, benign. Fructose appears in blood and urine (fructokinase deficiency is kinder).	Hypoglycemia, jaundice, cirrhosis, vomiting. Symptoms only present following consumption of fruit, juice, or honey.
ADDITIONAL REMARKS	Urine dipstick will be ⊖ (tests for glucose only); reducing sugar can be detected in the urine (nonspecific test for inborn errors of carbohydrate metabolism).	
TREATMENT	—	↓ intake of fructose, sucrose (glucose + fructose), and sorbitol (metabolized to fructose).



Disorders of galactose metabolism

	Galactokinase deficiency	Classic galactosemia
ENZYME DEFICIENCY	Galactokinase (autosomal recessive).	Galactose-1-phosphate uridylyltransferase (autosomal recessive).
PATHOPHYSIOLOGY	Galactitol accumulates if diet has galactose.	Damage caused by accumulation of toxic substances (eg, galactitol).
PRESENTATION (SIGNS/SYMPTOMS)	Relatively mild/benign condition (galactokinase deficiency is kinder). Galactose appears in blood (galactosemia) and urine (galactosuria); infantile cataracts. May present as failure to track objects or develop social smile.	Symptoms start when infant is fed formula or breast milk → failure to thrive, jaundice, hepatomegaly, infantile cataracts (galactitol deposition in eye lens), intellectual disability. Can predispose neonates to <i>E coli</i> sepsis.
TREATMENT	—	Exclude galactose and lactose (galactose + glucose) from diet.



Sorbitol

An alternative method of trapping glucose in the cell is to convert it to its alcohol counterpart, sorbitol, via aldose reductase. Some tissues then convert sorbitol to fructose using sorbitol dehydrogenase; tissues with an insufficient amount/activity of this enzyme are at risk of intracellular sorbitol accumulation, causing osmotic damage (eg, cataracts, retinopathy, and peripheral neuropathy seen with chronic hyperglycemia in diabetes).

High blood levels of galactose also result in conversion to the osmotically active galactitol via aldose reductase.

Liver, ovaries, and seminal vesicles have both enzymes (they **lose** sorbitol).



Lens has primarily Aldose reductase. Retina, Kidneys, and Schwann cells have only aldose reductase (**LARKS**).

Lactase deficiency

Insufficient lactase enzyme → dietary lactose intolerance. Lactase functions on the intestinal brush border to digest lactose (in milk and milk products) into glucose and galactose.

Primary: age-dependent decline after childhood (absence of lactase-persistent allele), common in people of Asian, African, or Native American descent.

Secondary: loss of intestinal brush border due to gastroenteritis (eg, rotavirus), autoimmune disease.

Congenital lactase deficiency: rare, due to defective gene.

Stool demonstrates ↓ pH and breath shows ↑ hydrogen content with lactose hydrogen breath test (H^+ is produced when colonic bacteria ferment undigested lactose). Intestinal biopsy reveals normal mucosa in patients with hereditary lactose intolerance.

FINDINGS

Bloating, cramps, flatulence (all due to fermentation of lactose by colonic bacteria → gas), and osmotic diarrhea (undigested lactose).

TREATMENT

Avoid dairy products or add lactase pills to diet; lactose-free milk.

Amino acids

Only L-amino acids are found in proteins.

Essential

PVT TIM HaLL: Phenylalanine, Valine, Tryptophan, Threonine, Isoleucine, Methionine, Histidine, Leucine, Lysine.

Glucogenic: Methionine, histidine, valine. We **met his valentine**, who is so **sweet** (glucogenic).

Glucogenic/ketogenic: Isoleucine, phenylalanine, threonine, tryptophan.

Ketogenic: leucine, lysine. The **only purely** ketogenic amino acids.

Acidic

Aspartic **acid**, glutamic **acid**.

Negatively charged at body pH.

Basic

Histidine, **lysine**, **arginine**.

Arginine is most **basic**. Histidine has no charge at body pH.

Arginine and histidine are required during periods of growth.

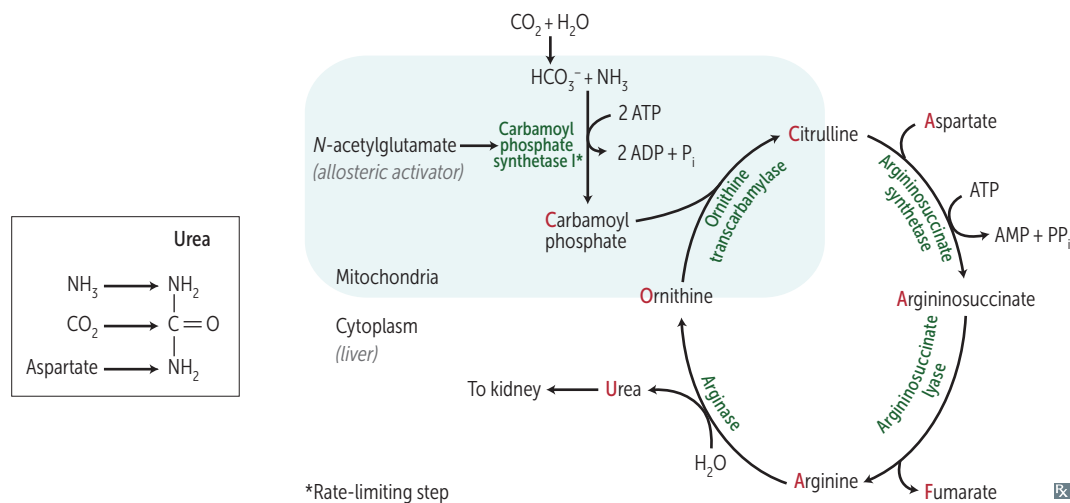
Arginine and lysine are ↑ in histones which bind negatively charged DNA.

His lys (lies) **are basic**.

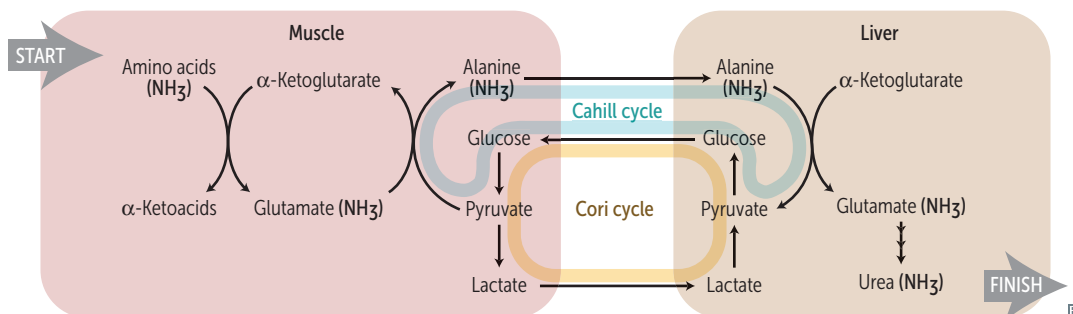
Urea cycle

Amino acid catabolism generates common metabolites (eg, pyruvate, acetyl-CoA), which serve as metabolic fuels. Excess nitrogen is converted to urea and excreted by the kidneys.

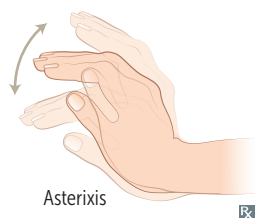
Ordinarily, Careless Crappers Are Also Frivolous About Urination.



Transport of ammonia by alanine



Hyperammonemia



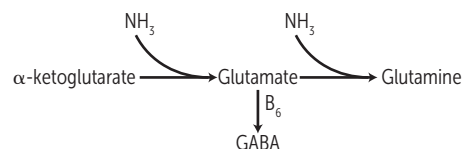
Can be acquired (eg, liver disease) or hereditary (eg, urea cycle enzyme deficiencies). Presents with flapping tremor (asterixis), slurring of speech, somnolence, vomiting, cerebral edema, blurring of vision. $\uparrow \text{NH}_3$ causes CNS toxicity, involving:

- TCA cycle inhibition ($\downarrow \alpha$ -ketoglutarate)
- $\downarrow$ glutamate
- $\uparrow$ GABAergic tone ($\uparrow$ GABA)
- $\uparrow$ glutamine
- Cerebral edema (glutamine induced osmotic shifts)

Treatment: limit protein in diet.

May be given to $\downarrow$ ammonia levels:

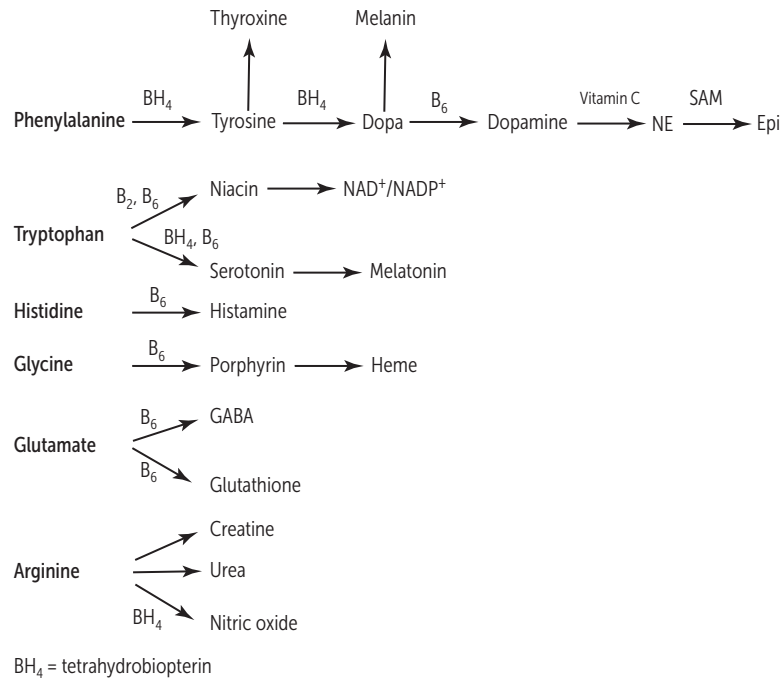
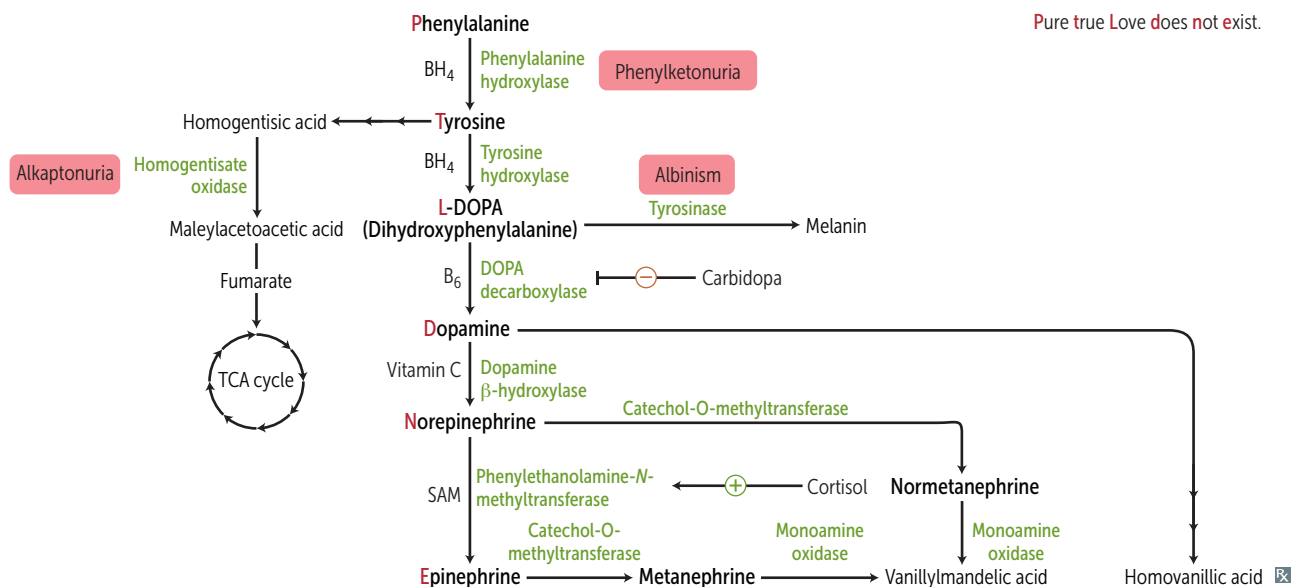
- Lactulose to acidify GI tract and trap NH_4^+ for excretion.
- Antibiotics (eg, rifaximin) to $\downarrow$ ammoniagenic bacteria.
- Benzoate, phenylacetate, or phenylbutyrate react with glycine or glutamine, forming products that are excreted renally.



Ornithine transcarbamylase deficiency

Most common urea cycle disorder. X-linked recessive (vs other urea cycle enzyme deficiencies, which are autosomal recessive). Interferes with the body's ability to eliminate ammonia. Often evident in the first few days of life, but may present later. Excess carbamoyl phosphate is converted to orotic acid (part of the pyrimidine synthesis pathway; vs. carbamoyl phosphate synthetase I deficiency).

Findings: ↑ orotic acid in blood and urine, ↓ BUN, symptoms of hyperammonemia. No megaloblastic anemia (vs orotic aciduria).

Amino acid derivatives**Catecholamine synthesis/tyrosine catabolism**

Phenylketonuria

Caused by ↓ phenylalanine hydroxylase (PAH). Tyrosine becomes essential. ↑ phenylalanine → ↑ phenyl ketones in urine.

Tetrahydrobiopterin (BH₄) deficiency—BH₄ essential cofactor for PAH. BH₄ deficiency → ↑ phenylalanine. Varying degrees of clinical severity. Untreated patients typically die in infancy.

Phenylalanine embryopathy—↑ phenylalanine levels in pregnant patients with untreated phenylketonuria (PKU) can cause fetal growth restriction, microcephaly, intellectual disability, congenital heart defects. Can be prevented with dietary measures.

Autosomal recessive.

Screening occurs 2–3 days after birth (normal at birth because of maternal enzyme during fetal life).

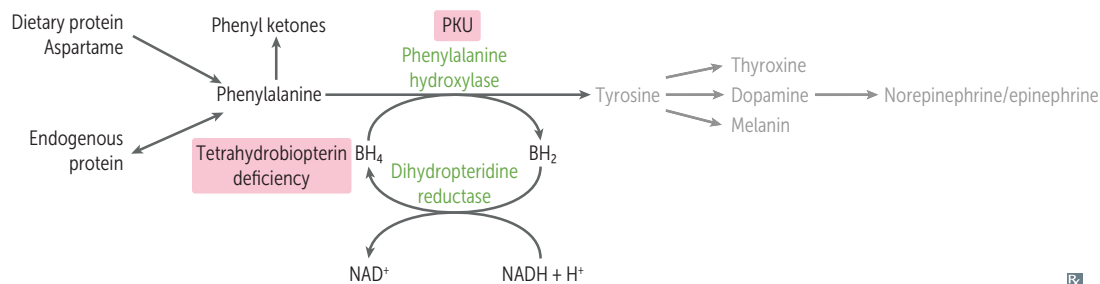
Findings: intellectual disability, microcephaly, seizures, hypopigmented skin, eczema, musty body odor. Findings are rare due to neonatal screening.

Treatment: ↓ phenylalanine and ↑ tyrosine in diet (eg, soy products, chicken, fish, milk), tetrahydrobiopterin supplementation.

Phenyl ketones—phenylacetate, phenyllactate, and phenylpyruvate.

Disorder of **aromatic** amino acid metabolism → musty body **odor**.

Patients with PKU must avoid the artificial sweetener aspartame, which is converted to phenylalanine.

**Maple syrup urine disease**

Blocked degradation of **branched** amino acids (**i**soleucine, **l**eucine, **v**aline) due to ↓ branched-chain α-ketoacid dehydrogenase (**B**₁). Causes ↑ α-ketoacids in the blood, especially those of leucine.

Treatment: restriction of isoleucine, leucine, valine in diet, and thiamine supplementation.

Autosomal recessive.

Presentation: vomiting, poor feeding, secretions (urine, sweat, ear wax) smell like maple syrup/burnt sugar. Causes progressive neurologic decline, including seizures and dystonia.

I love **V**ermont **m**aple **s**yrup from maple trees (with **B**₁**r**anches).

Alkaptonuria

Congenital deficiency of homogentisate oxidase in the degradative pathway of tyrosine to fumarate → pigment-forming homogentisic acid builds up in tissue. Autosomal recessive. Usually benign.

Findings: bluish-black connective tissue, ear cartilage, and sclerae (ochronosis **A**); urine turns black on prolonged exposure to air. May have debilitating arthralgias (homogentisic acid toxic to cartilage).

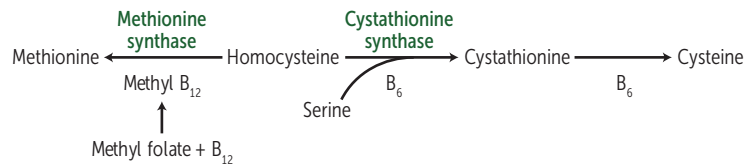
Homocystinuria

Causes (all autosomal recessive):

- Cystathionine synthase deficiency (treatment: ↓ methionine, ↑ cysteine, ↑ B₆, B₁₂, and folate in diet)
- ↓ affinity of cystathionine synthase for pyridoxal phosphate (treatment: ↑↑ B₆ and ↑ cysteine in diet)
- Methionine synthase (homocysteine methyltransferase) deficiency (treatment: ↑ methionine in diet)
- Methylene tetrahydrofolate reductase (MTHFR) deficiency (treatment: ↑ folate in diet)

All forms result in excess homocysteine.

HOMOCYstinuria: ↑↑ Homocysteine in urine, **O**steoporosis, **M**arfanoid habitus, **O**cular changes (downward and inward lens subluxation), **C**ardiovascular effects (thrombosis and atherosclerosis → stroke and MI), **kY**phosis, intellectual disability, hypopigmented skin. In homocystinuria, lens subluxes “down and in” (vs Mar**fan**, “up and **fan**s out”).

**Cystinuria**

Hereditary defect of renal PCT and intestinal amino acid transporter that prevents reabsorption of **C**ystine, **O**rnithine, **L**ysine, and **A**rginine (**COLA**).

Cystine is made of 2 cysteines connected by a disulfide bond.

Excess cystine in the urine can lead to recurrent precipitation of hexagonal cystine stones.

Treatment: urinary alkalinization (eg, potassium citrate, acetazolamide) and chelating agents (eg, penicillamine) ↑ solubility of cystine stones; good hydration; diet low in methionine.

Autosomal recessive. Common (1:7000).

Cystinuria detected with urinary sodium-cyanide nitroprusside test and proton nuclear magnetic resonance spectroscopy of urine.

Organic acidemias

Most commonly present in infancy with poor feeding, vomiting, hypotonia, high anion gap metabolic acidosis, hepatomegaly, seizures. Organic acid accumulation:

- Inhibits gluconeogenesis → ↓ fasting blood glucose levels, ↑ ketoacidosis → high anion gap metabolic acidosis
- Inhibits urea cycle → hyperammonemia

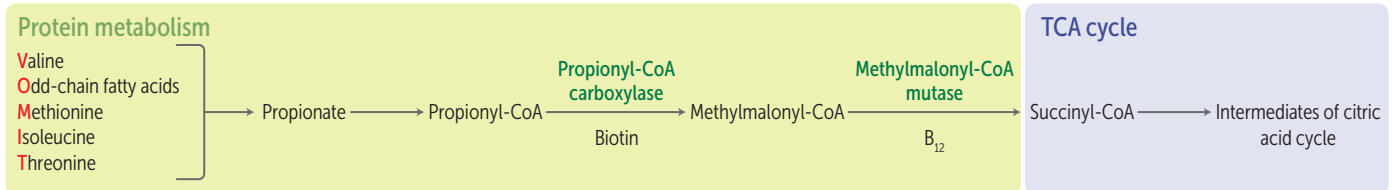
Propionic acidemia

Deficiency of propionyl-CoA carboxylase → ↑ propionyl-CoA, ↓ methylmalonic acid.

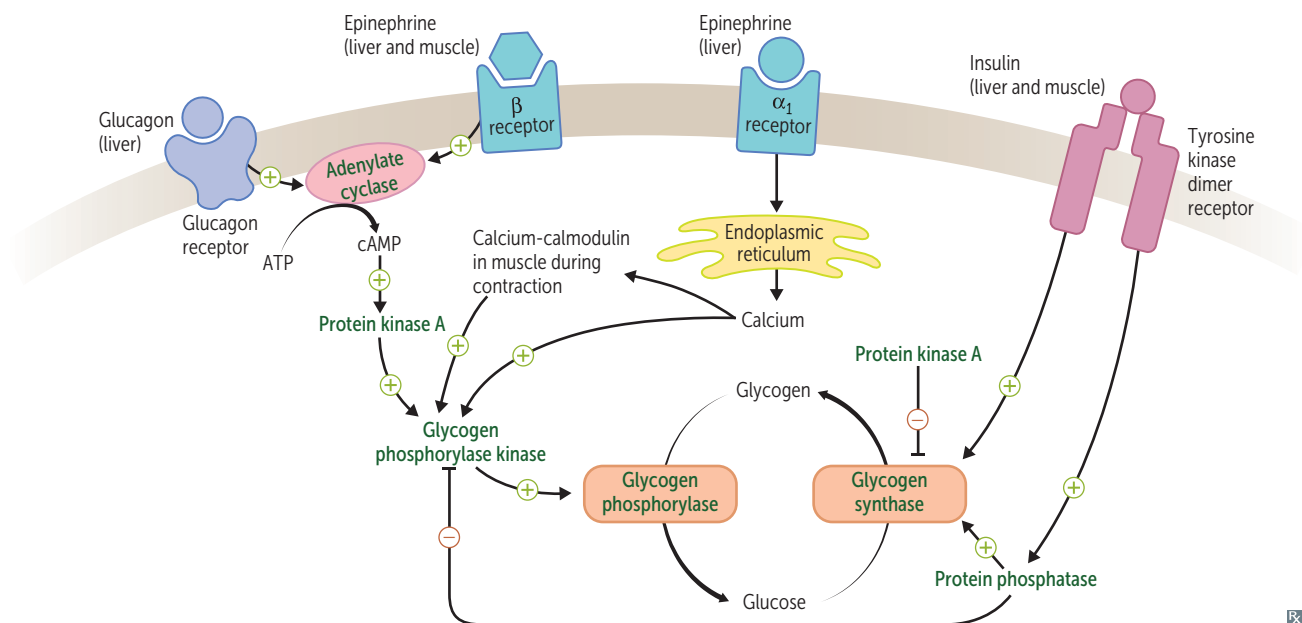
Treatment: low-protein diet limited in substances that metabolize into propionyl-CoA: **V**aline, **O**dd-chain fatty acids, **M**ethionine, **I**soleucine, **T**hreonine (**VOMIT**).

Methylmalonic acidemia

Deficiency of methylmalonyl-CoA mutase or vitamin B₁₂.



Glycogen regulation by insulin and glucagon/epinephrine



Glycogen

Branches have α -(1,6) bonds; linear linkages have α -(1,4) bonds.

Skeletal muscle

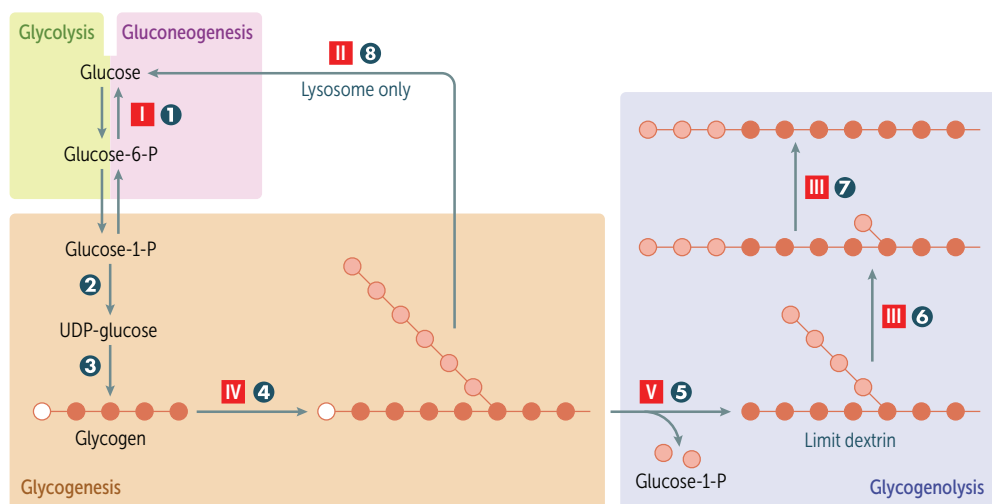
Glycogen undergoes glycogenolysis $\rightarrow$ glucose-1-phosphate $\rightarrow$ glucose-6-phosphate, which is rapidly metabolized during exercise.

Hepatocytes

Glycogen is stored and undergoes glycogenolysis to maintain blood sugar at appropriate levels.

Glycogen phosphorylase **5** liberates glucose-1-phosphate residues off branched glycogen until 4 glucose units remain on a branch. Then 4- α -D-glucanotransferase (debranching enzyme **6**) moves 3 of the 4 glucose units from the branch to the linear linkage. Then α -1,6-glucosidase (debranching enzyme **7**) cleaves off the last residue, liberating a free glucose.

Limit dextrin—2–4 residues remaining on a branch after glycogen phosphorylase has shortened it.



Glycogen storage disease type

- I** Von Gierke disease
- II** Pompe disease
- III** Cori disease
- IV** Anderson disease
- V** McArdle disease

Glycogen enzymes

- 1** Glucose-6-phosphatase
- 2** UDP-glucose pyrophosphorylase
- 3** Glycogen synthase
- 4** Branching enzyme
- 5** Glycogen phosphorylase
- 6** Debranching enzyme (4- α -D-glucanotransferase)
- 7** Debranching enzyme (α -1,6-glucosidase)
- 8** α -1,4-glucosidase

Note: A small amount of glycogen is degraded in lysosomes by **8** α -1,4-glucosidase (acid maltase).

Glycogen storage diseases

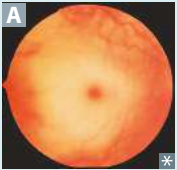
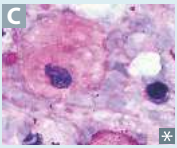
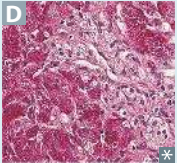
At least 15 types have been identified, all resulting in abnormal glycogen metabolism and an accumulation of glycogen within cells. Periodic acid–Schiff stain identifies glycogen and is useful in identifying these diseases.

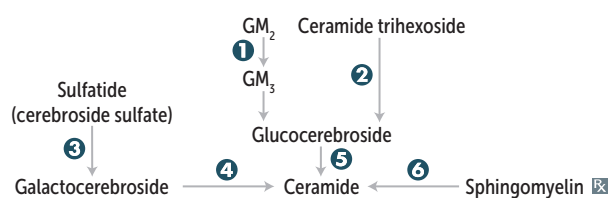
Vice **p**resident can't accept **m**oney.
Types I–V are autosomal recessive.
Andersen: **B**ranching.
Cori: **D**ebranching. (**A**BCD)

DISEASE	FINDINGS	DEFICIENT ENZYME	COMMENTS
Von Gierke disease (type I)	Severe fasting hypoglycemia, ↑↑ G lycogen in liver and kidneys, ↑ blood lactate, ↑ triglycerides, ↑ uric acid (G out), and hepatomegaly, renomegaly. Liver does not regulate blood glucose.	G lucose-6-phosphatase.	Treatment: frequent oral glucose/cornstarch; avoidance of fructose and galactose. Impaired gluconeogenesis and glycogenolysis.
Pompe disease (type II)	Cardiomyopathy, hypotonia, exercise intolerance, enlarged tongue, and systemic findings lead to early death.	Lysosomal acid α -1,4-glucosidase (acid maltase).	P ompe trashes the p ump (1st and 4th letter; heart, liver, and muscle).
Cori disease (type III)	Similar to von Gierke disease, but milder symptoms and normal blood lactate levels. Can lead to cardiomyopathy. Limit dextrin–like structures accumulate in cytosol; can lead to hepatomegaly, cirrhosis, and hepatic adenomas.	D ebranching enzymes (α -1,6-glucosidase and 4- α -D-glucanotransferase).	Gluconeogenesis is intact.
Andersen disease (type IV)	Most commonly presents with hepatosplenomegaly and failure to thrive in early infancy. Other findings include infantile cirrhosis, muscular weakness, hypotonia, cardiomyopathy early childhood death.	B ranching enzyme. Neuromuscular form can present at any age.	Hypoglycemia occurs late in the disease.
McArdle disease (type V)	↑ glycogen in muscle, but muscle cannot break it down → painful m uscle cramps, m yoglobinuria (red urine) with strenuous exercise, and arrhythmia from electrolyte abnormalities. Second-wind phenomenon noted during exercise due to ↑ muscular blood flow.	Skeletal muscle glycogen phosphorylase (m yophosphorylase). Characterized by a flat venous lactate curve with normal rise in ammonia levels during exercise.	Blood glucose levels typically unaffected. M cArdle = m uscle.

Lysosomal storage diseases

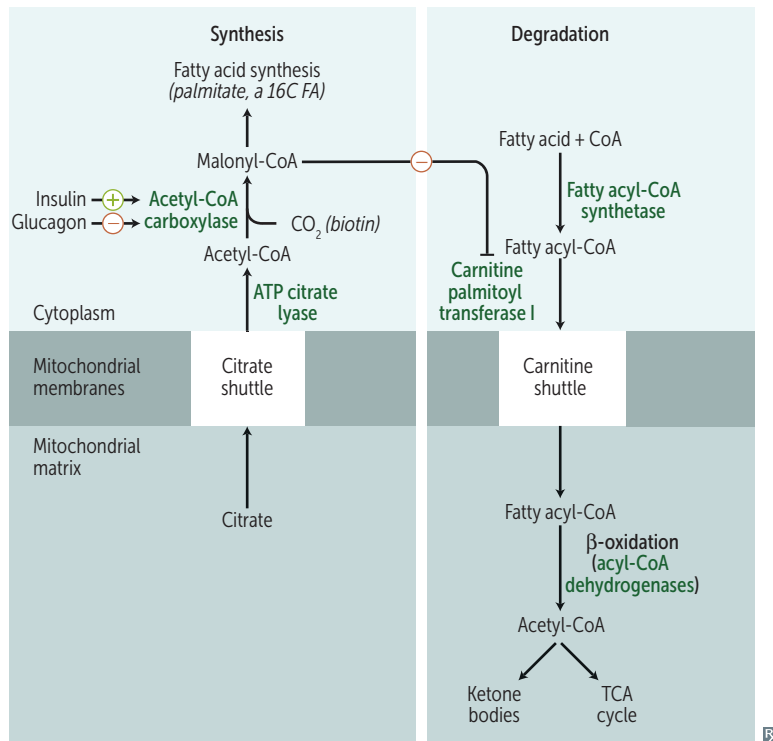
Lysosomal enzyme deficiency → accumulation of abnormal metabolic products. ↑ incidence of Tay-Sachs, Niemann-Pick, and some forms of Gaucher disease in Ashkenazi Jews.

DISEASE	FINDINGS	DEFICIENT ENZYME	ACCUMULATED SUBSTRATE	INHERITANCE
Sphingolipidoses				
Tay-Sachs disease 	Progressive neurodegeneration, developmental delay/regression, hyperreflexia, hyperacusis, “cherry-red” [*] spot on macula A (lipid accumulation in ganglion cell layer), lysosomes with onion skin, no hepatosplenomegaly.	1 Hexosaminidase A (“ T Ay- S ax”).	GM ₂ ganglioside.	AR
Fabry disease 	Early: triad of episodic peripheral neuropathy, angiokeratomas B , hypohidrosis. Late: progressive renal failure, cardiovascular disease.	2 α-galactosidase A; treat with recombinant α-galactosidase.	Ceramide trihexoside (globotriaosylceramide).	XR
Metachromatic leukodystrophy	Central and peripheral demyelination with ataxia, dementia.	3 Arylsulfatase A.	Cerebroside sulfate.	AR
Krabbe disease	Peripheral neuropathy, destruction of oligodendrocytes, developmental delay, CN II atrophy, globoid cells.	4 Galactocerebrosidase (galactosylceramidase).	Galactocerebroside, psychosine.	AR
Gaucher disease 	Most common. Hepatosplenomegaly, pancytopenia, osteoporosis, avascular necrosis of femur, bone crises, Gaucher cells (lipid-laden macrophages resembling crumpled tissue paper C).	5 Glucocerebrosidase (β-glucosidase); treat with recombinant glucocerebrosidase.	Glucocerebroside.	AR
Niemann-Pick disease 	Progressive neurodegeneration, hepatosplenomegaly (vs Tay-Sachs disease), foam cells (lipid-laden macrophages) D , “cherry-red” [*] spot on macula A .	6 Sphingomyelinase.	Sphingomyelin, cholesterol.	AR
Mucopolysaccharidoses				
Hurler syndrome	Developmental delay, hirsutism, skeletal anomalies, airway obstruction, clouded cornea, hepatosplenomegaly.	α-L-iduronidase.	Heparan sulfate, dermatan sulfate.	AR
Hunter syndrome	Mild Hurler + aggressive behavior, no corneal clouding.	Iduronate-2 (two)-sulfatase.		XR



Hunters see clearly (no corneal clouding) and aggressively aim for the **X** (X-linked recessive).

^{*}Red-tinted region at the center of the macula surrounded by retinal opacification.

Fatty acid metabolism

Fatty acid synthesis requires transport of citrate from mitochondria to cytosol. Predominantly occurs in liver, lactating mammary glands, and adipose tissue.

Long-chain fatty acid (LCFA) degradation requires carnitine-dependent transport into the mitochondrial matrix.

“**Sy**trate” = **syn**thesis.

Carnitine = **carn**age of fatty acids.

Systemic 1° carnitine deficiency—no cellular uptake of carnitine → no transport of LCFAs into mitochondria → toxic accumulation of LCFAs in the cytosol. Causes weakness, hypotonia, hypoketotic hypoglycemia, dilated cardiomyopathy.

Medium-chain acyl-CoA dehydrogenase deficiency—↓ ability to break down fatty acids into acetyl-CoA → accumulation of fatty acyl carnitines and dicarboxylic acids in the blood with hypoketotic hypoglycemia. Causes vomiting, lethargy, seizures, coma, liver dysfunction, hyperammonemia. Can lead to sudden death in infants or children. Treat by avoiding fasting.

Ketone bodies

In the liver, fatty acids and amino acids are metabolized to acetoacetate and β -hydroxybutyrate (to be used in muscle and brain).

In prolonged starvation and diabetic ketoacidosis, oxaloacetate is depleted for gluconeogenesis. With chronic alcohol overuse, high NADH state leads to accumulation of oxaloacetate (downregulated TCA cycle), shunting it to malate.

Ketone bodies: acetone (ketone), acetoacetate (ketoacid), β -hydroxybutyrate (ketoacid).
Breath smells like acetone (fruity odor).

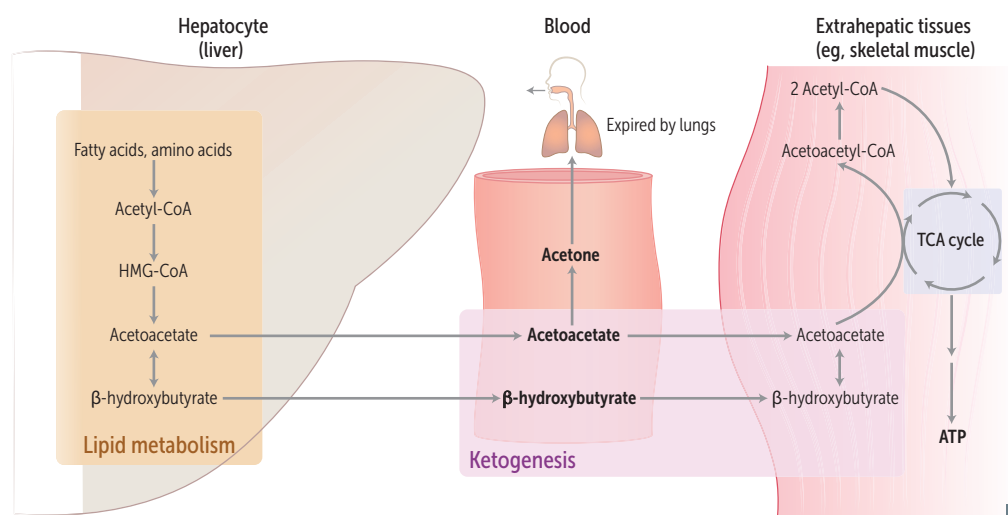
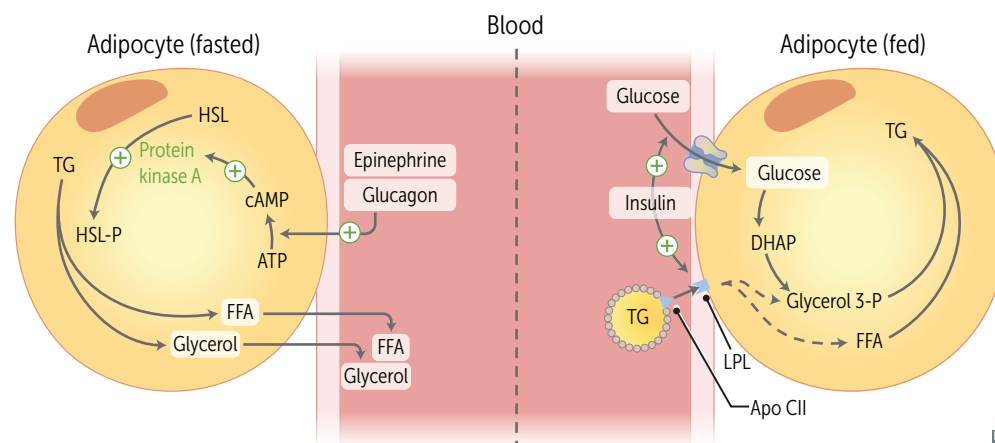
Urine test for ketones can detect acetoacetate, but not β -hydroxybutyrate.

RBCs cannot utilize ketone bodies; they strictly use glucose. Liver cells lack β ketoacyl-CoA transferase $\rightarrow$ cannot use ketone bodies as fuel.

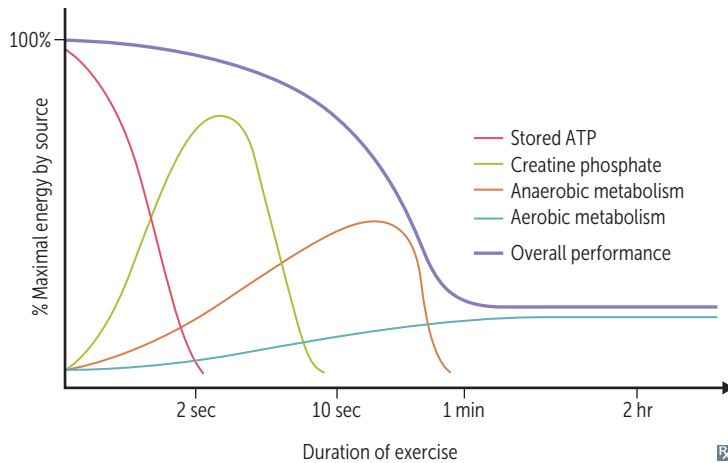
HMG-CoA lyase for ketone body production.

HMG-CoA reductase for cholesterol synthesis.

	Hyperammonemia	Hypoketosis	Ketosis
KETONE LEVELS	Normal	↓	↑
GLUCOSE LEVELS	Normal	↓	↓
DEFICIENCY	OTC (urea cycle)	MCAD deficiency	Methylmalonic acidemia, propionic acidemia

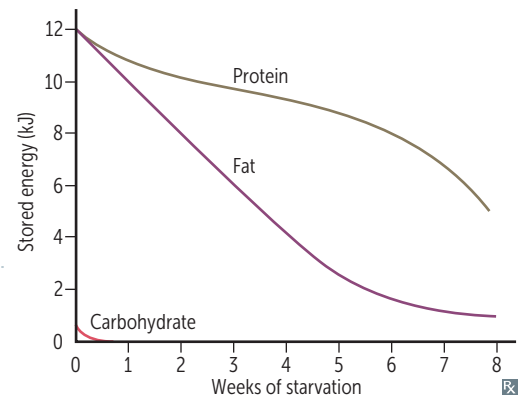
**Fasted vs fed state**

Metabolic fuel use

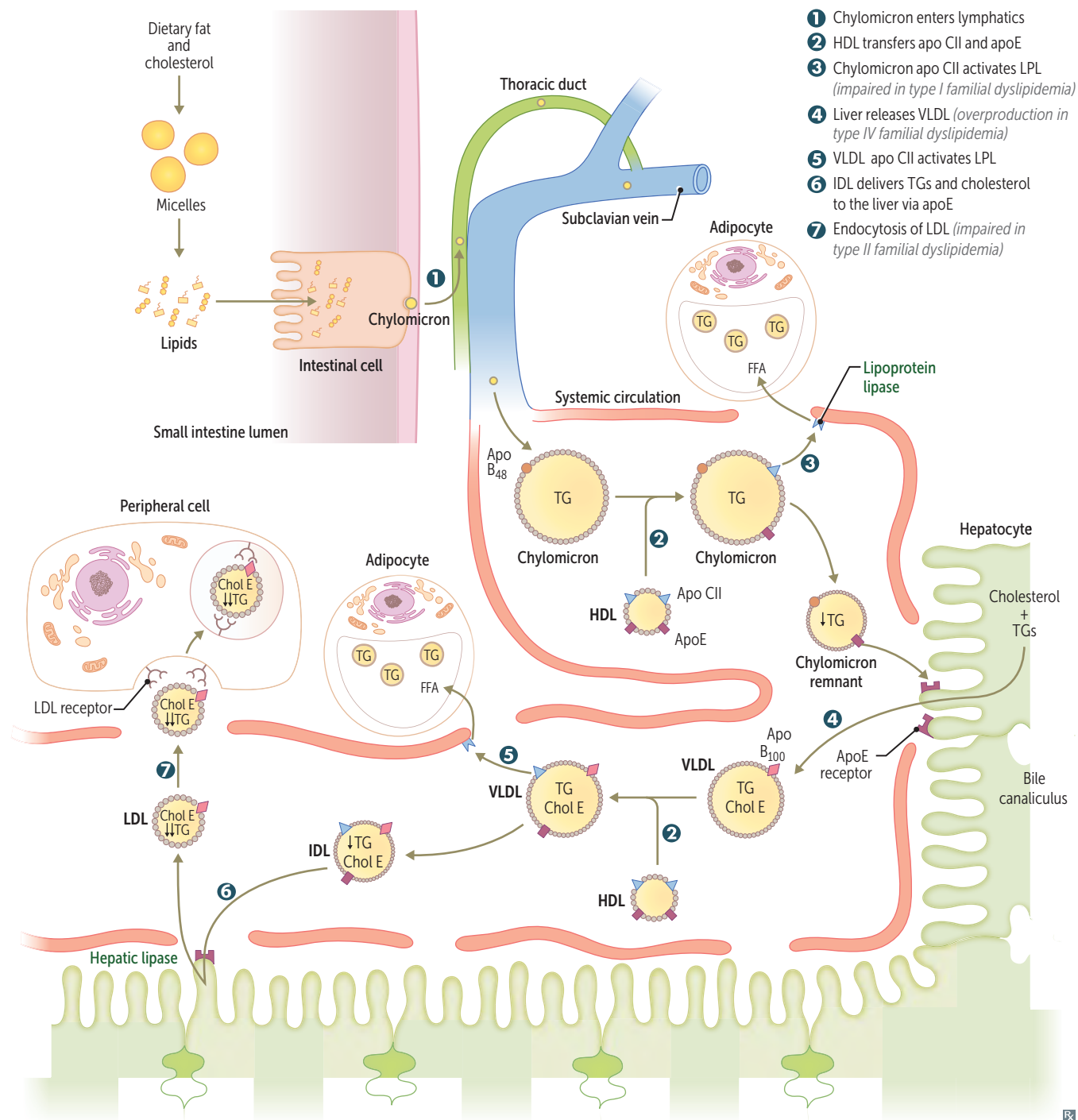


lg **carb**/protein = 4 kcal
lg **alcohol** = 7 kcal
lg **fatty acid** = 9 kcal
(# letters = # kcal)

Fasting and starvation	Priorities are to supply sufficient glucose to the brain and RBCs and to preserve protein.	
Fed state (after meals)	Glycolysis and aerobic respiration.	Insulin stimulates triglyceride (lipid) and glycogen (carbohydrate) storage alongside protein synthesis.
Fasting (between meals)	Hepatic glycogenolysis (major); hepatic gluconeogenesis, adipose release of FFA (minor).	Glucagon and epinephrine stimulate use of fuel reserves.
Starvation days 1–3	Blood glucose levels maintained by: ▪ Hepatic glycogenolysis ▪ Adipose release of FFA ▪ Muscle and liver, which shift fuel use from glucose to FFA ▪ Hepatic gluconeogenesis from peripheral tissue lactate and alanine, and from adipose tissue glycerol and propionyl-CoA (from odd-chain FFA—the only triacylglycerol component that contributes to gluconeogenesis)	Glycogen reserves depleted after day 1. RBCs lack mitochondria and therefore cannot use ketone bodies.
Starvation after day 3	Adipose stores (ketone bodies become the main source of energy for the brain). After these are depleted, vital protein degradation accelerates, leading to organ failure and death. Amount of excess stores determines survival time.	

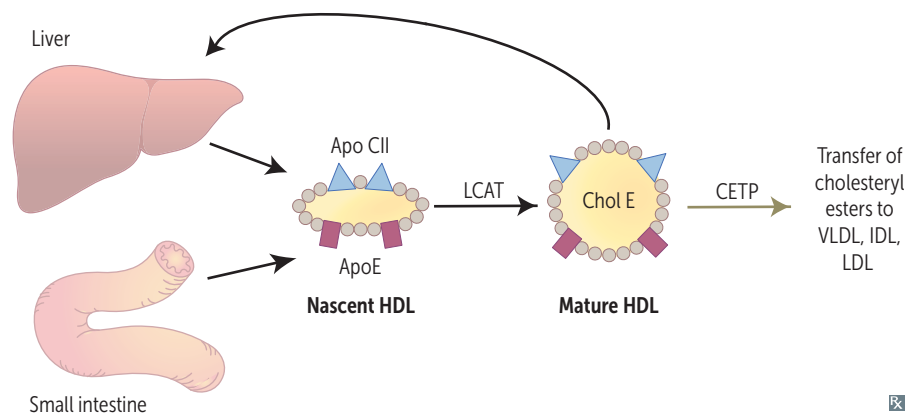


Lipid transport



Key enzymes in lipid transport

Cholesteryl ester transfer protein	Mediates transfer of cholesteryl esters to other lipoprotein particles.
Hepatic lipase	Degrades TGs remaining in IDL and chylomicron remnants.
Hormone-sensitive lipase	Degrades TGs stored in adipocytes. Promotes gluconeogenesis by releasing glycerol.
Lecithin-cholesterol acyltransferase	Catalyzes esterification of $\frac{2}{3}$ of plasma cholesterol (ie, required for HDL maturation).
Lipoprotein lipase	Degrades TGs in circulating chylomicrons and VLDL.
Pancreatic lipase	Degrades dietary TGs in small intestine.
PCSK9	Degrades LDL receptor → ↑ serum LDL. Inhibition → ↑ LDL receptor recycling → ↓ serum LDL.

**Major apolipoproteins**

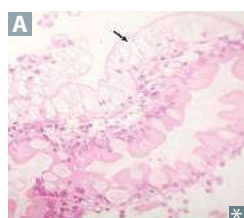
APOLIPOPROTEIN	FUNCTION	CHYLOMICRON	CHYLOMICRON REMNANT	VLDL	IDL	LDL	HDL
E	Mediates remnant uptake (everything e xcept LDL)	✓	✓	✓	✓		✓
A_I	Found only on a lpha-lipoproteins (HDL), a ctivates LCAT						✓
C_{II}	Lipoprotein lipase c ofactor that c atalyzes c leavage	✓		✓	✓		✓
B₄₈	Mediates chylomicron secretion into lymphatics Only on particles originating from the intestines	✓	✓				
B₁₀₀	Binds LDL receptor Only on particles originating from the l iver (I hope I l ive to B e 100)			✓	✓	✓	

Lipoprotein functions

Lipoproteins are composed of varying proportions of proteins, cholesterol, TGs, and phospholipids. LDL and HDL carry the most cholesterol.

Cholesterol is needed to maintain cell membrane integrity and synthesize bile acids, steroids, and vitamin D.

Chylomicron	Delivers dietary TGs to peripheral tissues. Delivers cholesterol to liver in the form of chylomicron remnants, which are mostly depleted of their TGs. Secreted by intestinal epithelial cells.
VLDL	Delivers hepatic TGs to peripheral tissue. Secreted by liver.
IDL	Delivers TGs and cholesterol to liver. Formed from degradation of VLDL.
LDL	Delivers hepatic cholesterol to peripheral tissues. Formed by hepatic lipase modification of IDL in the liver and peripheral tissue. Taken up by target cells via receptor-mediated endocytosis. LDL is L ethal.
HDL	Mediates reverse cholesterol transport from peripheral tissues to liver. Acts as a repository for apoC and apoE (which are needed for chylomicron and VLDL metabolism). Secreted from both liver and intestine. Alcohol ↑ synthesis. HDL is H ealthy.

Abetalipoproteinemia

Autosomal recessive. Mutation in gene that encodes microsomal transfer protein (*MTP*).

Chylomicrons, VLDL, LDL absent. Deficiency in apo B₄₈– and apo B₁₀₀–containing lipoproteins. Affected infants present with severe fat malabsorption, steatorrhea, failure to thrive. Later manifestations include retinitis pigmentosa, spinocerebellar degeneration due to vitamin E deficiency, progressive ataxia, acanthocytosis. Intestinal biopsy shows lipid-laden enterocytes (arrow in **A**).

Treatment: restriction of long-chain fatty acids, large doses of oral vitamin E.

Familial dyslipidemias

TYPE	INHERITANCE	PATHOGENESIS	↑ BLOOD LEVEL	CLINICAL
I—Hyper-chylomicronemia	AR	Lipoprotein lipase or apo CII deficiency	Chylomicrons, TG, cholesterol	Pancreatitis, hepatosplenomegaly, and eruptive/pruritic xanthomas (no ↑ risk for atherosclerosis). Creamy layer in supernatant.
II—Hyper-cholesterolemia	AD	Absent or defective LDL receptors, or defective apo B ₁₀₀	IIa: LDL, cholesterol IIb: LDL, cholesterol, VLDL	Heterozygotes (1:500) have cholesterol ≈ 300 mg/dL; homozygotes (very rare) have cholesterol ≥ 700 mg/dL. Accelerated atherosclerosis (may have MI before age 20), tendon (Achilles) xanthomas, and corneal arcus.
III—Dysbeta-lipoproteinemia	AR	Apo E (defective in type thr EE)	Chylomicrons, VLDL, TG	Premature atherosclerosis, tuberoeruptive and palmar xanthomas.
IV—Hyper-triglyceridemia	AD	Hepatic overproduction of VLDL	VLDL, TG	Hypertriglyceridemia (> 1000 mg/dL) can cause acute pancreatitis. Related to insulin resistance.